

Vikas Velagapudi

Atlanta, USA

+1-571-398-8989 | vikasvelagapudi57@gmail.com | [linkedin.com/in/vikas-velagapudi-48441a166/](https://www.linkedin.com/in/vikas-velagapudi-48441a166/)

Summary

Results-driven software engineer with three years of experience in developing ML models and backend processes. Proficient in Python, TensorFlow, PyTorch, and LLMs, with hands-on exposure to the complete ML pipeline from data preprocessing to model evaluation. Committed to innovation and collaborative development in agile environments, eager to contribute to generative AI initiatives and cross-functional projects.

Skills

- **Languages & Frameworks:** Python, C, SQL, JavaScript, SAP ABAP, PHP, HTML, CSS, PegaSystems
- **Python Libraries:** Pandas, NumPy, TensorFlow, PyTorch, Matplotlib, Scikit-learn
- **Other:** Machine Learning, Agile Methodologies, React, SEngineering, User Journey Mapping, Asset Maps, Natural Language Processing, Data Science, Large Language Models

Work Experience

Pelican IT Group

Jul 2025 - Present

Software Analyst

- Analyzed client software systems and documented detailed business and functional requirements to support custom IT solutions.
- Collaborated with cross-functional teams at both in-house and client locations to define system specifications and process improvements.
- Conducted comprehensive software analysis, including feasibility studies and impact assessments, to enhance enterprise software solutions.
- Supported the software development lifecycle (SDLC) by translating client needs into actionable technical requirements and use cases.

Mercor AI

Feb 2025 - Jun 2025

Computer Science Expert

- Executed AI research using the Feather.OpenAI platform to benchmark and refine machine learning models, emphasizing evaluation and improvement of model accuracy.
- Applied machine learning expertise and large-scale data processing techniques to evaluate AI-generated responses while leveraging Python and ML frameworks like TensorFlow and PyTorch.
- Utilized SQL for data manipulation and integrated ML training pipeline methodologies for thorough data preprocessing and model assessment.
- Gained practical experience in AI benchmarking, model evaluation, and research methodologies to support scalable ML solutions.

Entrepreneurs of Tomorrow

Sep 2024 - Jun 2025

Product Development Intern

- Designed and developed interactive websites and authoring tools for Learning Management System (LMS), enhancing user engagement.
- Collaborated with cross-functional teams to gather requirements and integrate web technologies, contributing to agile project management.
- Enhanced application performance and scalability through rigorous testing and debugging across multiple platforms.
- Accelerated development by leveraging low-code/no-code solutions, aligning with Agile principles, and refining UI/UX.

University Life - George Mason University

Jul 2023 - Jun 2024

Graduate Professional Assistant

- Ran for and was elected as President of the Graduate and Professional Student Association, advocating for graduate student interests.
- Contributed to policy changes that benefited graduate students by participating in decision-making bodies like the Board of Visitors, Graduate Council, and Provost Search Committee
- Managed the budget of GAPSA, ensuring fiscal responsibility and allocation of resources to support graduate initiatives.
- Chaired the General Assembly leading discussions among graduate reps to address concerns, propose and pass resolutions regarding graduate student life and academics

University Life - George Mason University

Feb 2023 - Jun 2023

Project Assistant

- Led initiatives as part of the UL PPM Team to improve university applications, enhancing user satisfaction by implementing targeted improvements
- Used Asset Maps and User Journeys to evaluate and improve user experiences, making applications of GMU more intuitive and user-friendly
- Created and conducted surveys to capture student feedback, identifying key issues and opportunities to refine user journeys, resulting in actionable insights for application improvements

- Generated comprehensive reports based on survey data, helping make informed decisions for application enhancements, leading to more effective and targeted improvements as part of SDLC processes

Deloitte USI

Jul 2021 - Aug 2022

Software Engineer (Analyst)

- Developed and optimized backend processes for the 1ERP project at Verizon, with a focus on bill payment automation, user contract setup, and financial reconciliation.
- Used SAP ABAP, SQL, and Programming concepts to deliver solutions that significantly improved data processing efficiency
- Programmed and implemented logical processes like bill payments, new user contracts, and Due clearance for Autopay users.
- Created a project scheduler and maintained daily task notes, streamlining progress tracking and team communication

Education

George Mason University

Aug 2022 - May 2024

Masters of Science, Computer Science, Minor in Machine Learning

- **GPA:** 3.6/4

KL University

Aug 2017 - May 2021

Bachelor of Technology, Software Engineering

- **GPA:** 9.1/10

Projects

EconBERTa-Towards Robust Extraction of Named Entities in Economics

- Re-developed and enhanced EconBERTa, an python-based LLM model tailored to economic texts, as part of a 3-member team to improve NER economic tasks.
- Pre-trained the model on the ECON-IE dataset, including 1,000 annotated abstracts focused on economic policy interventions.
- Achieved state-of-art results by surpassing models BERT, RoBERTa, and DeBERTa in NER tasks, improving accuracy and robustness.
- Extended EconBERTa to a multilingual version, enhancing its applicability and impact on economics evaluations across languages.

Sentiment Analysis on Movie Reviews (IMDb)

- Conducted sentiment analysis on a large dataset of over 3 GB, consisting of 5.5 million movie reviews from IMDb using python libraries PyTorch and TensorFlow.
- Implemented multiple machine learning models using python including L R, Linear SVC, Random Forest, and MLP to classify reviews as +/-
- Utilized the Helpfulness column to identify valuable reviewers in IMDb, enriching the depth of analysis and insights obtained.
- Investigated the impact of different rows on sentiment analysis accuracy, with r_detail proving to be more informative feature.

Resume Review Application

- Built a responsive web application using React for dynamic UI, allowing users to upload resumes and receive AI-driven feedback.
- Connected the app to Gemini AI via RESTful APIs to analyze and evaluate resumes based on industry standards and best practices.
- Designed the backend in Python with Flask to handle API requests and provide real-time processing of resume data.
- Streamlined user experience through asynchronous data handling and intuitive design, ensuring quick and accurate results.

Digital Outing Pass System for Hostellers

Digital Outing Pass System for Hostellers

- Developed and implemented a system aimed at streamlining the process of hostel outing request approvals for students. Using MySQL as the primary database
- Created a user-friendly website where students could digitally apply for outing passes, and track and view their outing status converting a physical paper trail into a end to end web application.
- Integrated ID card scanners at security gate for real-time verification of outing permissions, enhancing security measures.
- Managed project codebase and version control using GitHub and Linux, successfully deployed and currently used at KL University.

1ERP B2C Enhancements (Bill To Cash Enhancements)

- Contributed to the 1ERP Verizon project at Deloitte as an SAP ABAP developer, specializing in backend processes.
- Engaged in the development and enhancement of various reports and RICEF objects to align with client requirements.
- Programmed logical processes such as bill payment reports, new user contracts, and Due clearance for Autopay users.
- Successfully tested and delivered new enhancements, ensuring alignment with project specifications and client expectations.